

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5 FT	(a)			Missing state = liquid (1) Missing process = boiling / evaporation / vaporisation (1)	2			2		
	(b)			2 nd and 5 th boxes ticked i.e. The atoms in a solid are in fixed positions (1) A solid has atoms that vibrate more as they gain energy (1) Lose 1 mark for each extra tick	2			2		
	(c)	(i)		660 [°C]		1		1	1	
		(ii)		3 000 [s]		1		1	1	
	(d)	(i)		Substitution: $\frac{288\,000\,000}{3\,000}$ (1) = 96 000 [W] (1) Conversion = 96 [kW] (1) Alternative: Conversion = 288 000 [kJ] (1) $\frac{288\,000}{3\,000}$ (1) = 96 [kW] (1) N.B. Conversion by dividing by 1 000 award 1 mark	1	1 1		3	3	
		(ii)		Substitution: $\frac{288\,000\,000}{900 \times 640}$ or $\frac{288\,000\,000}{576\,000}$ (1) = 500 [kg] (1)	1	1		2	2	
	(e)			Substitution: $1\,500 \times 400\,000$ (1) = 600 000 000 [J] (1)	1	1		2	2	
				Question 5 total	7	6	0	13	9	0